Expno 5

Experiment on Packet capture tool: wireshark

Packet Sniffer:

- Sniffs messages being sent / received from by your computer
- Store and display the content of the various protocal field in the messages
- Passive program:
  - never sends packets itsef
  - no packet addressed to it
  - received a copy of all packet (send/received)

Packet Sniffer Structure Diagnostic Tools

- Tcp dump

  eg
  ```
  tcp dump -en x host 10.129.41.2 -w
                    exe 3.out
  ```

- wireshark

  ```
  wireshark -r exe3.out
  ```

Description

wireshark :

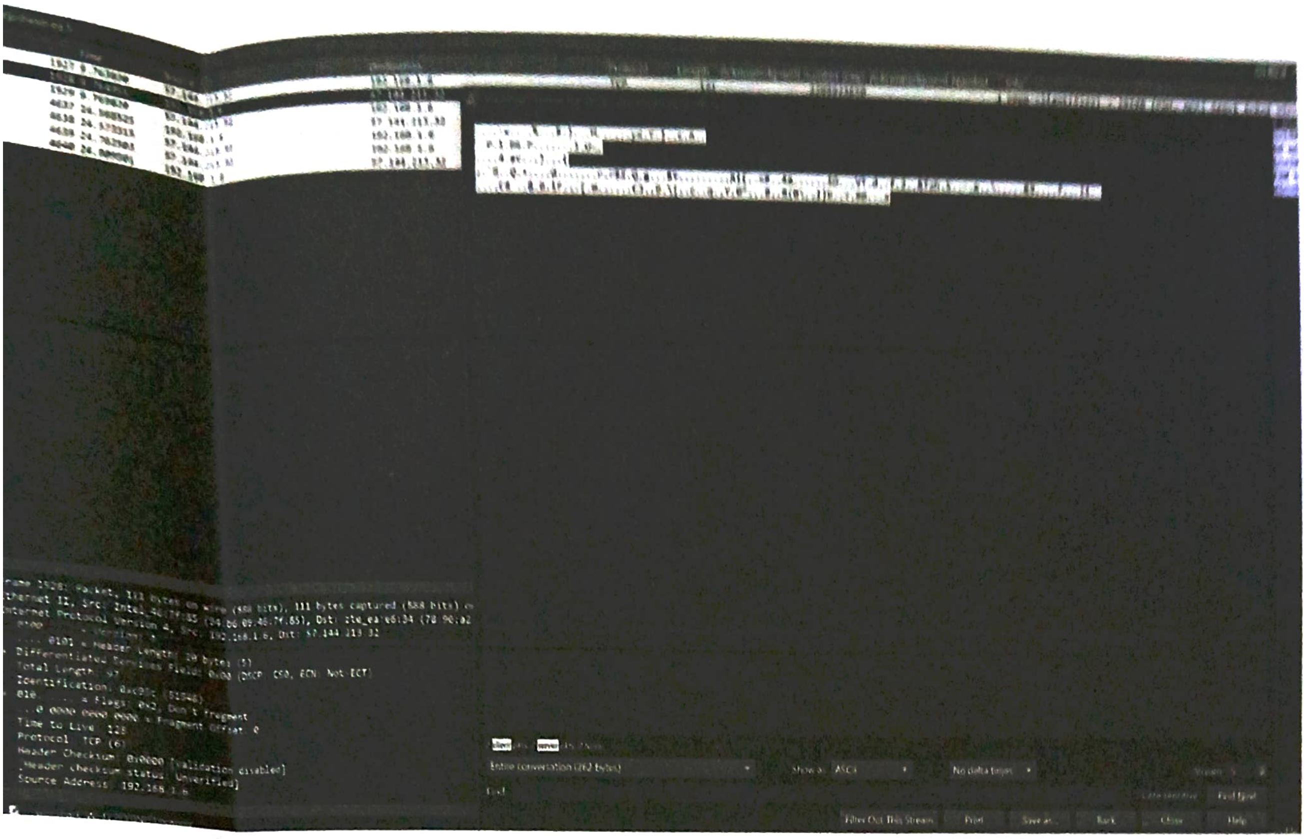
wireShark a network analysis tool formerl known as Ethereal, captures packet in real ti and display them in human-readable format wireshark includes filters, colour coding and ot features that let you dig deep into network traff and inspect individual packets.

What we can do with wireshark

- Capture network traffic
- Decode packet protocol using dissectors
- Define filter - capture and display
- Watch smart statistics
- Analyze problems

Wireshark used for :

- Network adminstrater : troubleshoot networ problems.
- Network security engineer : escamines securit problems
- Developers : debug protocol implementatio

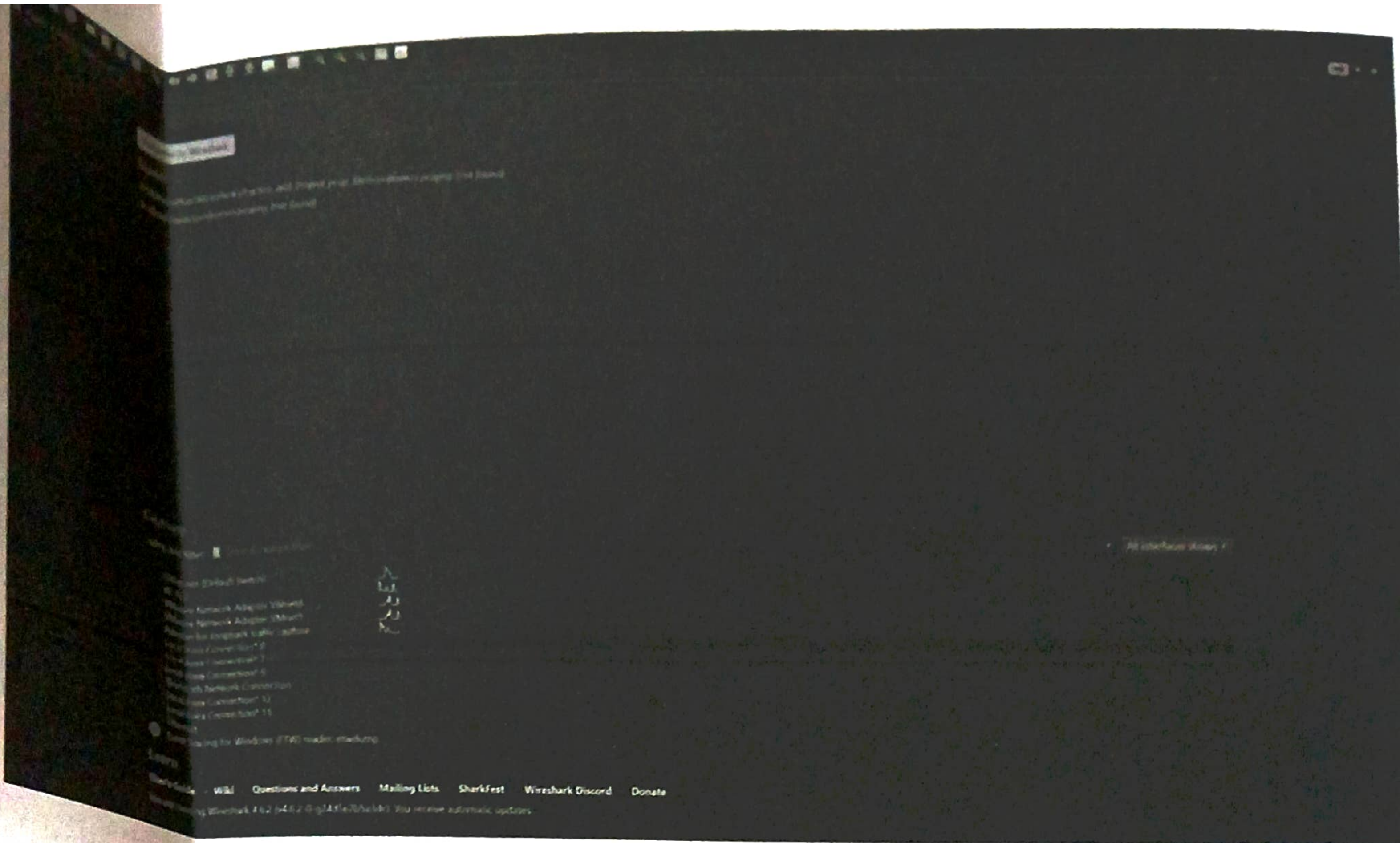


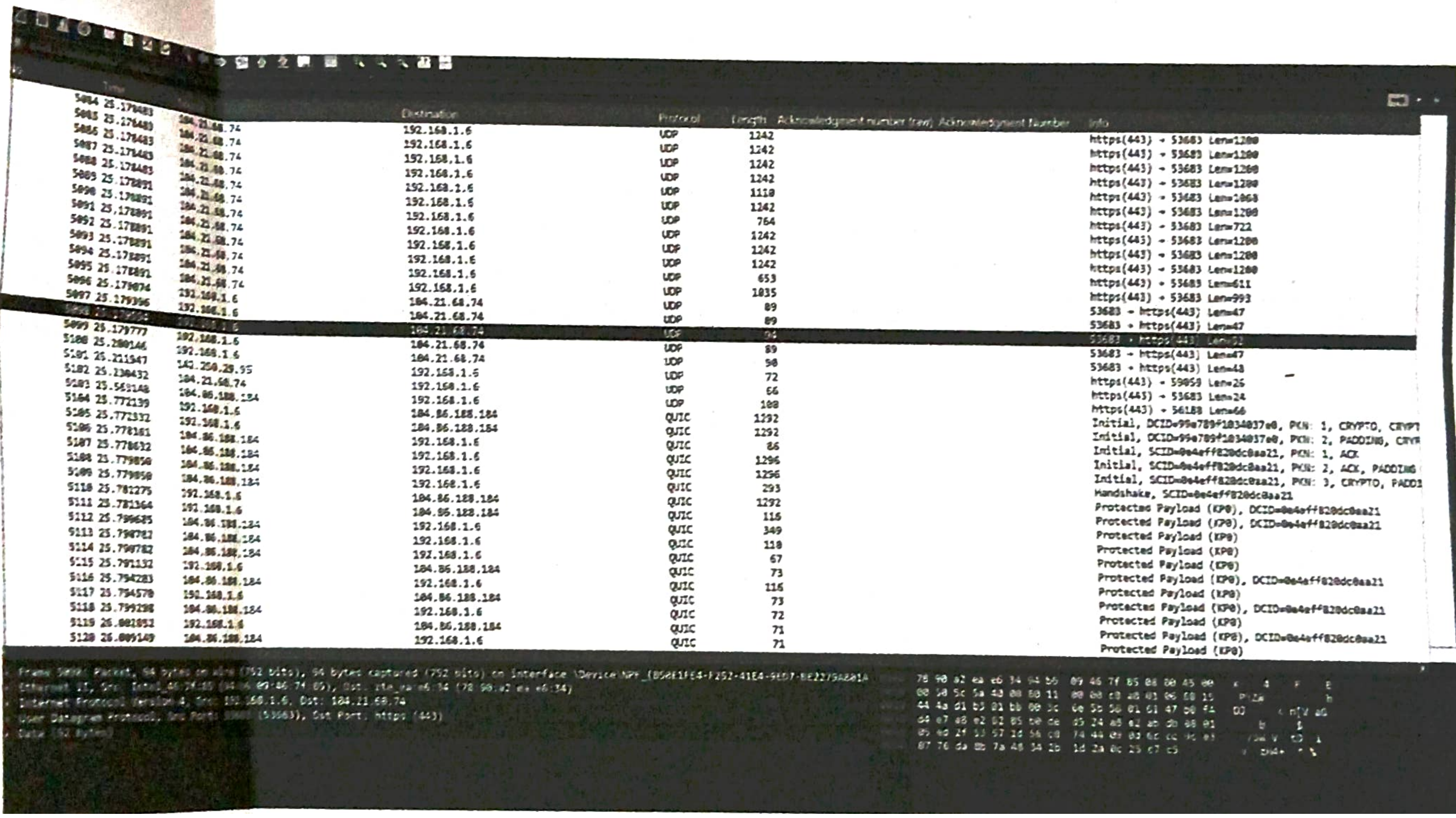
```
rame 1528: Packet, 111 bytes on wire (888 bits), 111 bytes captured (888 bits) o
thernet II, Src: Intel_46:7f:85 (94:b6:09:46:7f:85), Dst: zte_ea:e6:34 (78:90:a2
nternet Protocol Version 4, Src: 192.168.1.6, Dst: 57.144.213.32
  0100 .... = Version: 4
  .... 0101 = Header Length: 20 bytes (5)
> Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
  Total Length: 97
  Identification: 0xc95c (51548)
> 010. .... = Flags: 0x2, Don't fragment
  ...0 0000 0000 0000 = Fragment Offset: 0
  Time to Live: 128
  Protocol: TCP (6)
  Header Checksum: 0x0000 [validation disabled]
  [Header checksum status: Unverified]
  Source Address: 192.168.1.6
```

Entire conversation (262 bytes)

Show as: ASCII

No delta times





```
No.  Time       Source          Destination     Protocol  Length  Acknowledgment number (raw)  Acknowledgment Number  Info
5084 25.178483  104.21.68.74    192.168.1.6     UDP       1242    https(443) → 53683 Len=1200
5085 25.178483  104.21.68.74    192.168.1.6     UDP       1242    https(443) → 53683 Len=1200
5086 25.178483  104.21.68.74    192.168.1.6     UDP       1242    https(443) → 53683 Len=1200
5087 25.178483  104.21.68.74    192.168.1.6     UDP       1242    https(443) → 53683 Len=1200
5088 25.178483  104.21.68.74    192.168.1.6     UDP       1110    https(443) → 53683 Len=1068
5089 25.178891  104.21.68.74    192.168.1.6     UDP       1242    https(443) → 53683 Len=1200
5090 25.178891  104.21.68.74    192.168.1.6     UDP       764     https(443) → 53683 Len=722
5091 25.178891  104.21.68.74    192.168.1.6     UDP       1242    https(443) → 53683 Len=1200
5092 25.178891  104.21.68.74    192.168.1.6     UDP       1242    https(443) → 53683 Len=1200
5093 25.178891  104.21.68.74    192.168.1.6     UDP       1242    https(443) → 53683 Len=1200
5094 25.178891  104.21.68.74    192.168.1.6     UDP       653     https(443) → 53683 Len=611
5095 25.178891  104.21.68.74    192.168.1.6     UDP       1035    https(443) → 53683 Len=993
5096 25.179074  192.168.1.6     104.21.68.74    UDP       89      53683 → https(443) Len=47
5097 25.179396  192.168.1.6     104.21.68.74    UDP       89      53683 → https(443) Len=47
5098 [illegible] 192.168.1.6    104.21.68.74    UDP       94      53683 → https(443) Len=52
5099 25.179777  192.168.1.6     104.21.68.74    UDP       89      53683 → https(443) Len=47
5100 25.200146  192.168.1.6     104.21.68.74    UDP       90      53683 → https(443) Len=48
5101 25.211947  142.250.29.95   192.168.1.6     UDP       72      https(443) → 59059 Len=26
5102 25.230432  104.21.68.74    192.168.1.6     UDP       66      https(443) → 53683 Len=24
5103 25.569148  104.86.188.184  192.168.1.6     UDP       108     https(443) → 56188 Len=66
5104 25.772139  192.168.1.6     104.86.188.184  QUIC      1292    Initial, DCID=99e789f1034037e0, PKN: 1, CRYPTO, CRYPT
5105 25.772332  192.168.1.6     104.86.188.184  QUIC      1292    Initial, DCID=99e789f1034037e0, PKN: 2, PADDING, CRYP
5106 25.778161  104.86.188.184  192.168.1.6     QUIC      86      Initial, SCID=0e4eff820dc0aa21, PKN: 1, ACK
5107 25.778632  104.86.188.184  192.168.1.6     QUIC      1296    Initial, SCID=0e4eff820dc0aa21, PKN: 2, ACK, PADDING
5108 25.779050  104.86.188.184  192.168.1.6     QUIC      1296    Initial, SCID=0e4eff820dc0aa21, PKN: 3, CRYPTO, PADDI
5109 25.779050  104.86.188.184  192.168.1.6     QUIC      293     Handshake, SCID=0e4eff820dc0aa21
5110 25.782275  192.168.1.6     104.86.188.184  QUIC      1292    Protected Payload (KP0), DCID=0e4eff820dc0aa21
5111 25.783364  192.168.1.6     104.86.188.184  QUIC      116     Protected Payload (KP0), DCID=0e4eff820dc0aa21
5112 25.790685  104.86.188.184  192.168.1.6     QUIC      349     Protected Payload (KP0)
5113 25.790782  104.86.188.184  192.168.1.6     QUIC      110     Protected Payload (KP0)
5114 25.790782  104.86.188.184  192.168.1.6     QUIC      67      Protected Payload (KP0)
5115 25.791132  192.168.1.6     104.86.188.184  QUIC      73      Protected Payload (KP0), DCID=0e4eff820dc0aa21
5116 25.794283  104.86.188.184  192.168.1.6     QUIC      116     Protected Payload (KP0)
5117 25.794570  192.168.1.6     104.86.188.184  QUIC      73      Protected Payload (KP0), DCID=0e4eff820dc0aa21
5118 25.799298  104.86.188.184  192.168.1.6     QUIC      72      Protected Payload (KP0)
5119 26.001853  192.168.1.6     104.86.188.184  QUIC      71      Protected Payload (KP0), DCID=0e4eff820dc0aa21
5120 26.009149  104.86.188.184  192.168.1.6     QUIC      71      Protected Payload (KP0)
```

```
> Frame 5098: Packet, 94 bytes on wire (752 bits), 94 bytes captured (752 bits) on interface \Device\NPF_{B50E1FE4-F252-41E4-9ED7-8E2279A8A01A
> Ethernet II, Src: Intel_46:7f:85 ([illegible]:46:7f:85), Dst: zte_ea:e6:34 (78:90:a2:ea:e6:34)
> Internet Protocol Version 4, Src: 192.168.1.6, Dst: 104.21.68.74
> User Datagram Protocol, Src Port: [illegible] (53683), Dst Port: https (443)
> Data (52 bytes)

78 90 a2 ea e6 34 94 b5  09 46 7f 85 08 00 45 00
00 50 5c 5a 40 00 80 11  00 00 c0 a8 01 06 68 15
44 4a d1 b3 01 bb 00 3c  6e 5b 50 01 61 47 b0 f4
d4 e7 a8 e2 02 05 b0 de  d5 24 a5 e2 ab db 08 01
05 ed 2f 13 57 1d 56 c8  74 44 d9 03 6c cc 9c e1
07 76 da 0b 7a 48 34 2b  1d 2a 0c 25 c7 c5
```

Getting Wireshark

Wireshark can be downloaded for windo or macOS from its official website for Linux another UNIX-like system, Wireshark will be found in its packag repositaries. for Ubuntu, wireshark will be found in the Ubuntu S Centre.

Capturing Packets:

After downloading and installing Wiresh lanch it and double-click the name of a netw interface and capture to start capturing pack on the interface

As Soon as you click the interface's name, sell the packet start to appear in real time wire Shark captures each packet sent to or form your System.

8M Result: wireshark help us in analysing and understanding our network traffic and packet flow through various filters and flow graph